MIT ACADEMY OF ENGINEERING

DEPARTMENT OF CSE AIML

ASSIGNMENT 3

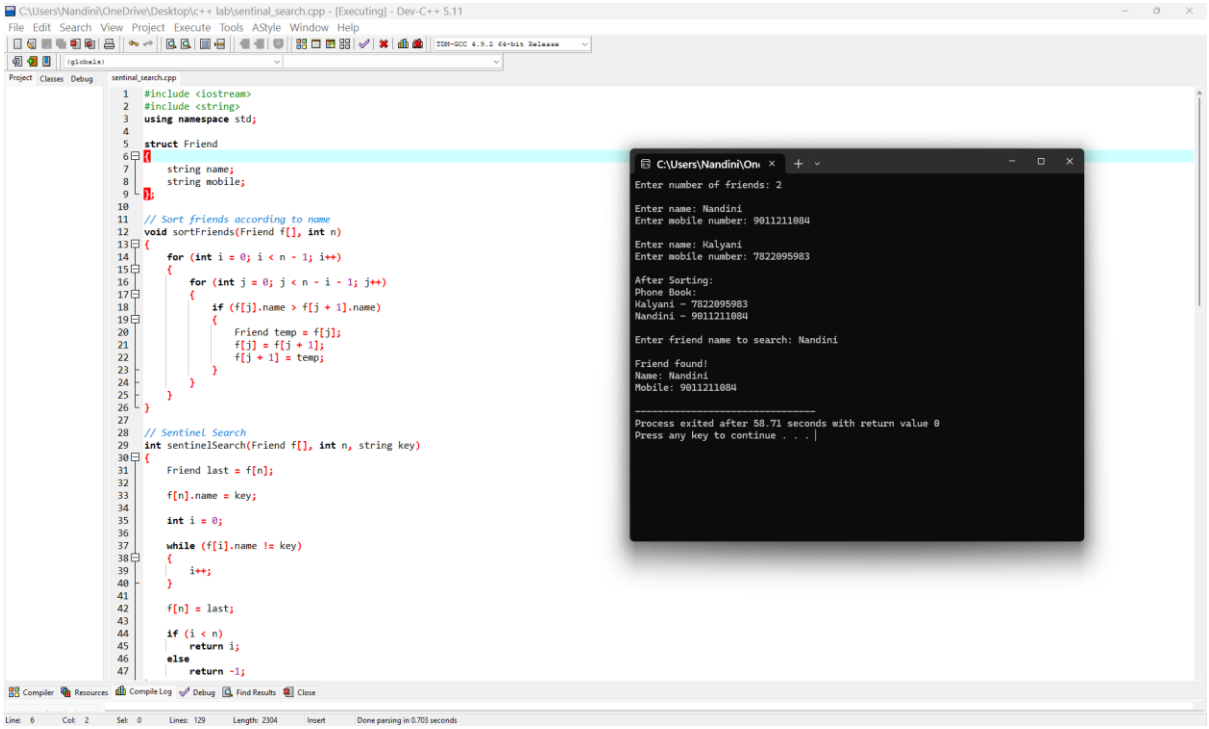
Name : Nandini Ravindra Chimandare

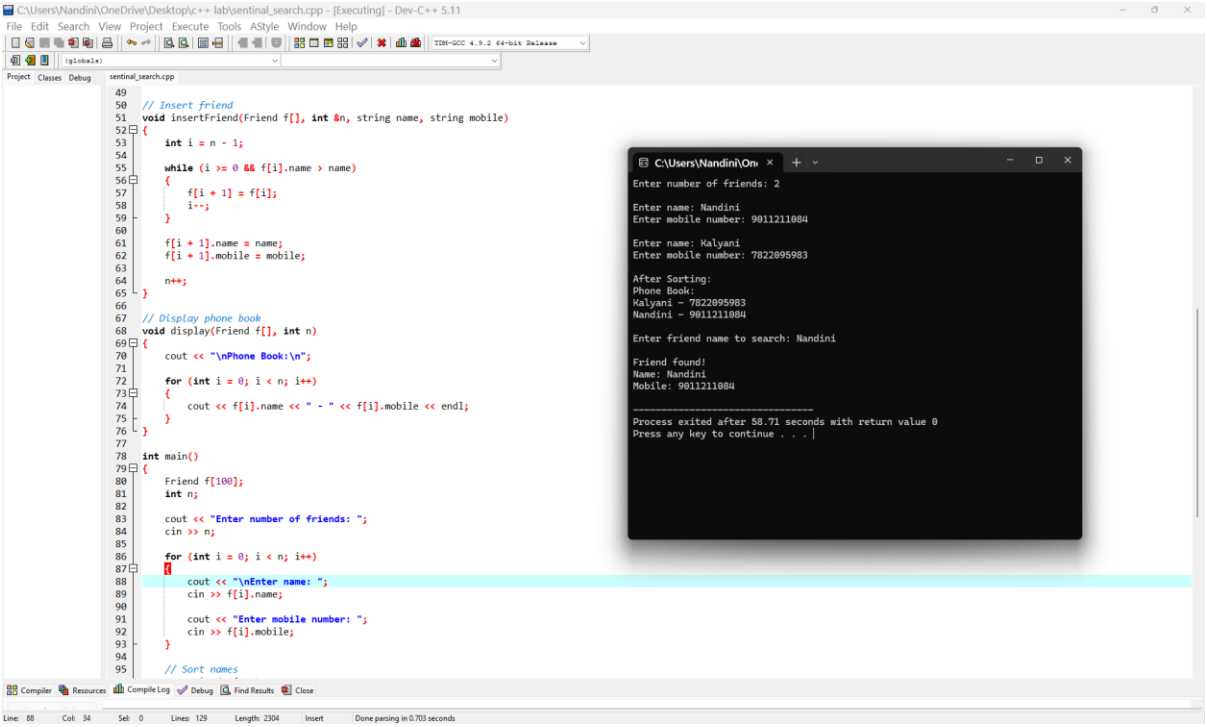
Div/Batch : AIML B1

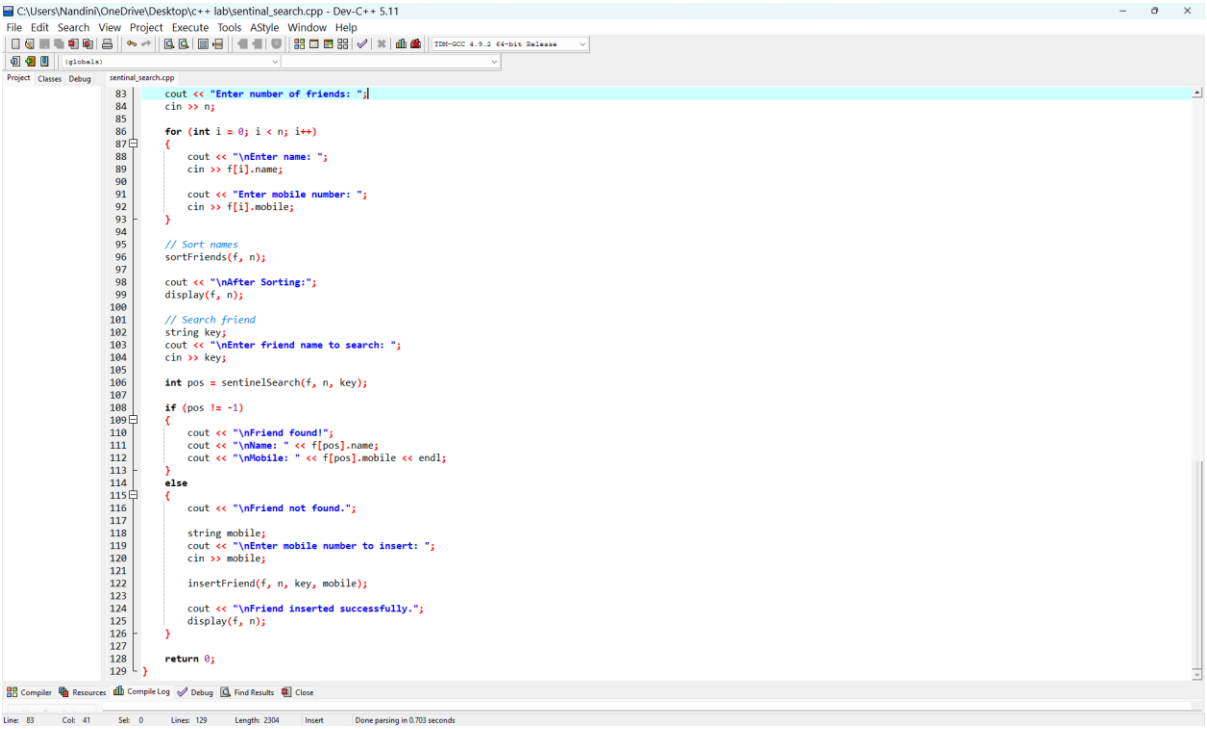
PRN : 202501110108

Topic : Sentinal Search

Implementation:




C:\Users\Nandini\OneDrive\Desktop\c++ lab\sentinal_search.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
sentinal_search.cpp
// Insert friend
void insertFriend(Friend f[], int &n, string name, string mobile)
{
int i = n - 1;
while (i >= 0 && f[i].name > name)
{
f[i + 1] = f[i];
i--;
}
f[i + 1].name = name;
f[i + 1].mobile = mobile;
n++;
}
// Display phone book
void display(Friend f[], int n)
{
cout << "\nPhone Book:\n";
for (int i = 0; i < n; i++)
{
cout << f[i].name << " - " << f[i].mobile << endl;
}
}
int main()
{
Friend f[100];
int n;
cout << "Enter number of friends: ";
cin >> n;
for (int i = 0; i < n; i++)
{
cout << "\nEnter name: ";
cin >> f[i].name;
cout << "Enter mobile number: ";
cin >> f[i].mobile;
}
// Sort names
C:\Users\Nandini\On
Enter number of friends: 2
Enter name: Nandini
Enter mobile number: 9011211084
Enter name: Kalyani
Enter mobile number: 7822095983
After Sorting:
Phone Book:
Kalyani - 7822095983
Nandini - 9011211084
Enter friend name to search: Nandini
Friend found!
Name: Nandini
Mobile: 9011211084
Process exited after 58.71 seconds with return value 0
Press any key to continue . . .
Compiler Resources Compile Log Debug Find Results Close
Line: 88 Col: 34 Sel: 0 Lines: 129 Length: 2304 Insert Done parsing in 0.703 seconds


C:\Users\Nandini\OneDrive\Desktop\c++ lab\sentinal_search.cpp - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
sentinal_search.cpp
cout << "Enter number of friends: ";
cin >> n;
for (int i = 0; i < n; i++)
{
cout << "\nEnter name: ";
cin >> f[i].name;
cout << "Enter mobile number: ";
cin >> f[i].mobile;
}
// Sort names
sortFriends(f, n);
cout << "\nAfter Sorting:";
display(f, n);
// Search friend
string key;
cout << "\nEnter friend name to search: ";
cin >> key;
int pos = sentinelSearch(f, n, key);
if (pos != -1)
{
cout << "\nFriend found!";
cout << "\nName: " << f[pos].name;
cout << "\nMobile: " << f[pos].mobile << endl;
}
else
{
cout << "\nFriend not found.";
string mobile;
cout << "\nEnter mobile number to insert: ";
cin >> mobile;
insertFriend(f, n, key, mobile);
cout << "\nFriend inserted successfully.";
display(f, n);
}
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Line: 83 Col: 41 Sel: 0 Lines: 129 Length: 2304 Insert Done parsing in 0.703 seconds

Code:

```
#include <iostream>
#include <string>
using namespace std;

struct Friend
{
   string name;
   string mobile;
};

// Sort friends according to name
void sortFriends(Friend f[], int n)
{
   for (int i = 0; i < n - 1; i++)
   {
      for (int j = 0; j < n - i - 1; j++)
      {
         if (f[j].name > f[j + 1].name)
         {
            Friend temp = f[j];
            f[j] = f[j + 1];
            f[j + 1] = temp;
         }
      }
   }
}

// Sentinel Search
int sentinelSearch(Friend f[], int n, string key)
{
```

```
    Friend last = f[n];

    f[n].name = key;

    int i = 0;

    while (f[i].name != key)
    {
        i++;
    }

    f[n] = last;

    if (i < n)
        return i;
    else
        return -1;
}

// Insert friend
void insertFriend(Friend f[], int &n, string name, string mobile)
{
    int i = n - 1;

    while (i >= 0 && f[i].name > name)
    {
        f[i + 1] = f[i];
        i--;
    }

    f[i + 1].name = name;
```

```
    f[i + 1].mobile = mobile;

    n++;
}

// Display phone book
void display(Friend f[], int n)
{
    cout << "\nPhone Book:\n";

    for (int i = 0; i < n; i++)
    {
        cout << f[i].name << " - " << f[i].mobile << endl;
    }
}

int main()
{
    Friend f[100];
    int n;

    cout << "Enter number of friends: ";
    cin >> n;

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter name: ";
        cin >> f[i].name;

        cout << "Enter mobile number: ";
        cin >> f[i].mobile;
```

```
}

// Sort names
sortFriends(f, n);

cout << "\nAfter Sorting:";
display(f, n);

// Search friend
string key;
cout << "\nEnter friend name to search: ";
cin >> key;

int pos = sentinelSearch(f, n, key);

if (pos != -1)
{
   cout << "\nFriend found!";
   cout << "\nName: " << f[pos].name;
   cout << "\nMobile: " << f[pos].mobile << endl;
}
else
{
   cout << "\nFriend not found.";

   string mobile;
   cout << "\nEnter mobile number to insert: ";
   cin >> mobile;

   insertFriend(f, n, key, mobile);
```

```
        cout << "\nFriend inserted successfully.";
        display(f, n);
    }

    return 0;
}
```